ENERGY PLANNING FOR INDIA

Decision Variables

	Solar	Wind	Hydro	Coal	Gas
Uttar Pradesh	22000	2000	500	5400	23023.52941
Maharashtra	15000	5000	1000	4800	28164.70588
Tamil Nadu	10000	12000	500	2400	14905.88235
Gujarat	30000	15000	200	2700	10581.17647
Rajasthan	40000	8000	200	2100	7435.294118
Madhya Pradesh	10000	3000	500	2550	12182.35294
Karnataka	12000	6000	2000	1200	10494.11765
Andhra Pradesh	12000	8000	500	2250	7611.764706
Himachal Pradesh	800	400	4000	9.13047E-13	9.411764706
Assam	500	200	1500	150	2076.470588

Objective Function

Minimize total cost	8376500000	3874000000	872000000	1059750000	3494541176	17676791176
Cost of installation	55,000	65,000	80,000	45,000	30,000	

Constraints

							LHS		RHS
Capacity and Demand	Uttar Pradesh	0.2	0.1	0.1	0.7	0.85	28000	>=	28,000
	Maharashtra	0.22	0.25	0.15	0.7	0.85	32000	>=	32,000
	Tamil Nadu	0.2	0.3	0.1	0.7	0.85	20000	>=	20,000
	Gujarat	0.23	0.28	0.08	0.7	0.85	22000	>=	22,000
	Rajasthan	0.25	0.15	0.05	0.7	0.85	19000	>=	19,000
	Madhya Pradesh	0.22	0.2	0.12	0.7	0.85	15000	>=	15,000
	Karnataka	0.22	0.25	0.3	0.7	0.85	14500	>=	14,500
	Andhra Pradesh	0.22	0.28	0.15	0.7	0.85	13000	>=	13,000
	Himachal Pradesh	0.18	0.12	0.45	0.7	0.85	2000	>=	2,000
	Assam	0.17	0.1	0.35	0.7	0.85	2500	>=	2,500
Coal ceiling cap	Uttar Pradesh	22000	2000	500	5400	23023.52941	5,400	<=	21,169
	Maharashtra	15000	5000	1000	4800	28164.70588	4,800	<=	21,586
	Tamil Nadu	10000	12000	500	2400	14905.88235	2,400	<=	15,922
	Gujarat	30000	15000	200	2700	10581.17647	2,700	<=	23,392
	Rajasthan	40000	8000	200	2100	7435.294118	2,100	<=	23,094
	Madhya Pradesh	10000	3000	500	2550	12182.35294	2,550	<=	11,293
	Karnataka	12000	6000	2000	1200	10494.11765	1,200	<=	12,678
	Andhra Pradesh	12000	8000	500	2250	7611.764706	2,250	<=	12,145
	Himachal Pradesh	800	400	4000	9.13047E-13	9.411764706	0	<=	2,084
	Assam	500	200	1500	150	2076.470588	150	<=	1,771
Coal floor minimum	Uttar Pradesh				5400		5400	>=	5,400
	Maharashtra				4800		4800	>=	4,800
	Tamil Nadu				2400		2400	>=	2,400
	Gujarat				2700		2700	>=	2,700
	Rajasthan				2100		2100	>=	2,100
	Madhya Pradesh				2550		2550	>=	2,550
	Karnataka				1200		1200	>=	1,200
	Andhra Pradesh				2250		2250	>=	2,250
	Himachal Pradesh				9.13047E-13		9.13047E-13	>=	0
	Assam				150		150	>=	150
Grid Stability via conventional sources	Uttar Pradesh			50	5400	23023.52941	7000	<=	28473.52941
	Maharashtra			150	4800	28164.70588	8000	<=	33114.70588
	Tamil Nadu			50	2400	14905.88235	5000	<=	17355.88235
	Gujarat			16	2700	10581.17647	5500	<=	13297.17647
	Rajasthan			10	2100	7435.294118	4750	<=	9545.294118
	Madhya Pradesh			60	2550	12182.35294	3750	<=	14792.35294
	Karnataka			600	1200	10494.11765	3625	<=	12294.11765
	Andhra Pradesh			75	2250	7611.764706	3250	<=	9936.764706
	Himachal Pradesh			1800	9.13047E-13	9.411764706	500	<=	1809.411765
	Assam			525	150	2076.470588	625	<=	2751.470588
Renewables floor minimum									
For solar	Uttar Pradesh	22000					22000	>=	22,000
	Maharashtra	15000					15000	>=	15,000
	Tamil Nadu	10000					10000	>=	10,000
	Gujarat	30000					30000	>=	30,000
	Rajasthan	40000					40000	>=	40,000
	Madhya Pradesh	10000					10000	>=	10,000
	Karnataka	12000					12000	>=	12,000
	Andhra Pradesh	12000					12000	>=	12,000
	Himachal Pradesh	800					800	>=	800
	Assam	500					500	>=	500
For wind	Uttar Pradesh		2000				2000	>=	2,000
	Maharashtra		5000				5000	>=	5,000
	Tamil Nadu		12000				12000	>=	12,000
	Gujarat		15000				15000	>=	15,000
	Rajasthan		8000				8000	>=	8,000
	Madhya Pradesh		3000				3000	>=	3,000
	Karnataka		6000				6000	>=	6,000
	Andhra Pradesh		8000				8000	>=	8,000
	Himachal Pradesh		400				400	>=	400
	Assam		200				200	>=	200
For Hydro	Uttar Pradesh			500			500	>=	500
	Maharashtra			1000			1000	>=	1,000
	Tamil Nadu			500			500	>=	500
	Gujarat			200			200	>=	200
	Rajasthan			200			200	>=	200
	Madhya Pradesh			500			500	>=	500
	Karnataka			2000			2000	>=	2,000
	Andhra Pradesh			500			500	>=	500
	Himachal Pradesh			4000			4000	>=	4,000
	Assam			1500			1500	>=	1,500
Land Resource Limit	Uttar Pradesh	40,000					22,000	<=	40,000
	Maharashtra	30,000					15,000	<=	30,000
	Tamil Nadu	20,000					10,000	<=	20,000
	Gujarat	40,000					30,000	<=	40,000
	Rajasthan	75,000					40,000	<=	75,000
	Madhya Pradesh	35,000					10,000	<=	35,000
	Karnataka	25,000					12,000	<=	25,000
	Andhra Pradesh	30,000					12,000	<=	30,000
	Himachal Pradesh	2,000					800	<=	2,000
	Assam	3,000					500	<=	3,000
	Uttar Pradesh		5,000				2,000	<=	5,000
	Maharashtra		10,000				5,000	<=	10,000
	Tamil Nadu		35,000				12,000	<=	35,000
	Gujarat		30,000				15,000	<=	30,000
	Rajasthan		10,000				8,000	<=	10,000
	Madhya Pradesh		8,000				3,000	<=	8,000
	Karnataka		15,000				6,000	<=	15,000
	Andhra Pradesh		20,000				8,000	<=	20,000
	Himachal Pradesh		1,000				400	<=	1,000
	Assam		1,000				200	<=	1,000